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2 MARKS

Qno1: How GMOs are economically beneficial?

GMOs are an effective way to provide farmers a larger profit while spending less time resources. Economically they are beneficial because they are used to repel pests, which prevent the need for pesticides to be used.

Qno2: Define Vaccine and give an example.

A vaccine is a biological preparation that improves immunity to a particular disease. A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe, for example Varicella (chickenpox).

Qno3: What is Matrix Assisted Laser Desorption/Ionization?

MALDI is the abbreviation for "Matrix Assisted Laser Desorption/Ionization. " The sample for MALDI is uniformly mixed in a large quantity of matrix. The matrix absorbs the ultraviolet light (nitrogen laser light, wavelength 337 nm) and converts it to heat energy. A small part of the matrix (down to 100 nm from the top outer surface of the Analyte in the diagram) heats rapidly (in several nano seconds) and is vaporized, together with the sample.

3 MARKS

Qno1: Define Alcoholic fermentation.

• Done by

• Yeast and certain bacteria

• Pyruvate is broken into alcohol and carbon dioxide

• Used in the production of beer, wine and bread

Qno2: Define DNA vaccine and give example.

Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans.

DNA vaccines take immunization to a new technological level.

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These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material.

Example: Influenza vaccine

Qno3: Define Antimicrobial.

Antimicrobial products kill or slow the spread of microorganisms. Microorganisms include bacteria, viruses, protozoans, and fungi such as mold and mildew.

Antibacterials:

• Antibacterials are used to treat bacterial infections

• The drug toxicity to humans and other animals from antibacterials is generally considered low

• Antibacterials are among the most commonly used drugs and among the drugs commonly misused by physicians, for example, in viral respiratory tract infectionsly low.

• Prolonged use of certain antibacterials can decrease the number of gut flora which may have a negative impact on health

Antifungals:

• Antifungals are used to kill or prevent further growth of fungi.

• In medicine, they are used as a treatment for infections such as athlete's foot, ringworm and thrush and work by exploiting differences between mammalian and fungal cells.

• They kill off the fungal organism without dangerous effects on the host.

Antivirals:

• Antiviral drugs are a class of medication used specifically for treating viral infections.

• Like antibiotics, specific antivirals are used for specific viruses.

• They are relatively harmless to the host and therefore can be used to treat infections.

Antiparasitics:

• Antiparasitics are a class of medications indicated for the treatment of infection by parasites, such as nematodes, cestodes, trematodes, infectious protozoa, and amoebae.

• Like all antimicrobials against intracellular microbes, they must kill the infecting pest without serious damage to the host

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5 MARKS

Qno1: Describe Subunits Vaccines and give example.

• Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system.

• In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to immunization.

• Because subunit vaccines contain only the essential antigens and not all the other molecules that make up the microbe.

Example: Plague immunization.

Qno2: Write down the advantages of biopesticides.

• Inherently less harmful and less environmental load,

• Designed to affect only one specific pest or, in some cases, a few target organisms,

• Often effective in very small quantities and often decompose quickly, thereby resulting in lower exposures and largely avoiding the pollution problems.

• When used as a component of Integrated Pest Management (IPM) programs, biopesticides can contribute greatly.

Qno3: Describe in detail the importance of microbial biotechnology.

• Biotechnology is generally defined as application of living systems based technologies to develop commercial processes and products.

• Over the last few decades, several fundamental discoveries in life sciences have given rise to Modern Biotechnology which is now one of the fastest growing areas of science; hence this century has rightly been termed as 'Century of Biology'

• Agricultural biotechnology offers efficient and cost-effective means to produce a diverse array of novel, value-added products and tools.

• It has the potential to increase food production, reduce the dependency of agriculture on chemicals, and lower the cost of raw materials, all in an environmentally friendly manner.

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10 MARKS

Qno1: Explain Xanthomonas a plant pathogen.

Xanthan gum was discovered in the mid-1950s, during a systematic search for useful biopolymers. In a screen of its large microbial culture collection, the Northern Utilization Research and Development Division of the U.S.Department of Agriculture discovered that the bacterium Xanthomonas campestris (originally isolated fromthe rutabaga plant) produced a polysaccharide with potentially valuable physical properties. Substantial commercial production of xanthan gum began in 1964, and in 1969 the U.S. FDA authorized its use in food. Today, xanthan gum has numerous uses, in the food industry and elsewhere.The total U.S. consumption of xanthan in 2005 is estimated at more than 80 million pounds.

The genus Xanthomonas belongs to the γ subdivision of the proteobacteria and consists exclusively of plant pathogens. It is a relative of the genus Pseudomonas, a group of bacteria discussed elsewhere in this book in the context of the degradation of xenobiotics.

X. campestris pathovar (abbreviated “pv.”) campestris, the producer of xanthan gum, causes black rot, one of the most serious diseases of plants in the genus Brassica, which includes vegetables such as cabbage, cauliflower, brussels sprouts, broccoli, rutabaga, and turnips. Seeds contaminated with bacteria spread the infection. As the seedling emerges and grows, the bacteria colonize the surface of the plant.

In the case of X. campestris pv. campestris, the epiphytic bacteria enter the internal tissues through the hydathodes, structures on the leaf margin that allow the plant to excrete water. They then migrate through the vascular system and progressively cause chlorosis (destruction of chlorophyll), vein blackening (deposition of melaninlike pigments), and, ultimately, rotting.

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